

3810 Network security Syllabus

Course Prerequisites

- C– or better in **INFOTC 2810** OR
- C– or better in **CMP_SC 4850**

COURSE INFORMATION

This course is offered in seat. Attendance is required

- **Lecture Schedule:** Mon, Wed, and Fri 3:00PM - 3:50 PM.
- **Location:** Geological Sci Building 108.

Attendance Policy:

- In-seat students are required to attend lectures in person.
- Attendance will be taken

Course Description

This course introduces students to the principles, tools, and techniques of network security. Students will learn to design and implement secure network environments, configure network devices, apply security protocols, and troubleshoot network security issues. Emphasis will be placed on both theoretical foundations and practical, hands-on experience with up-to-date standardized systems and diagnostic tools.

Learning Outcomes

By the end of the course, students will be able to:

1. Identify and analyze network security threats and vulnerabilities.
2. Implement security measures on network devices and operating systems.

3. Configure and secure network infrastructure using industry-standard tools.
 4. Apply cryptographic techniques for data protection.
 5. Deploy and manage firewalls, IDPS, NAC, VPNs, and wireless security measures.
 6. Evaluate and implement network security policies, risk management, and disaster recovery plans.
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Course Topics

- Introduction to Network Security
 - Cryptography
 - Virtual Environment Configuration
 - Network Device Communication
 - Static Routing
 - RIPv1
 - RIPv2
 - EIGRP
 - OSPF
 - IS-IS
 - BGP
 - Setting Line Passwords
 - Subnetting – FLSM & VLSM
 - VLANs
 - Security in Routing and Switching
 - Inter-VLAN Routing
 - Network Access Control (NAC)
 - Firewalls – Windows, Linux, pfSense, Cisco Zone-Based, Access Control Lists
 - Intrusion Detection and Prevention Systems (IDPS)
 - Wireless Network Security
 - VPNs – Windows, Linux
 - Web Security
 - Network Security Management
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Course Schedule (15 Weeks)

Week 1 – Foundations

- Mon: Course introduction, syllabus, importance of network security
- Wed: Threats, vulnerabilities, and attacks
- Fri: Common network attacks (DoS, MITM, phishing, malware)

Week 2 – Cryptography

- Mon: Encryption and decryption; symmetric vs. asymmetric
- Wed: Hashing, digital signatures, certificates
- Fri: SSL/TLS, IPsec; Lab: encryption & hashing tools

Week 3 – Network Tools & Monitoring

- Mon: Packet inspection & network troubleshooting with built-in tools
- Wed: Network scanning & reconnaissance tools
- Fri: : Monitoring & Scanning Networks (CLI Tools Only)

Week 4 – Network Communication & Routing

Basics

- Mon: Physical/logical addressing
- Wed: Routing protocols overview
- Fri: Static routing theory

Week 5 – Routing Decisions and Metrics

- Mon: Administrative Distance (AD)
- Wed: EIGRP (Composite Metric)
- Fri: Advanced OSPF & Comparison

Week 6 – Load Balancing and Redundancy

- Mon: Load Balancing Basics (OSPF Focus)
- Wed: EIGRP Load Balancing
- Fri: Troubleshooting & Integration Case Study

Week 7 – Subnetting

- Mon: FLSM concepts and calculations

- Wed: VLSM concepts and calculations
- Fri: : subnetting & router configuration

Week 8 – VLANs & Security in Routing/Switching

- Mon: VLAN concepts & configuration
- Wed: VLAN
- Fri: VLAN security, securing router/switch configs (**Midterm – In-Class**)

Week 9 – Inter-VLAN Routing & NAC

- Mon: Inter-VLAN routing theory
- Wed: Inter-VLAN / PVLAN
- Fri: NAC theory – AAA, authentication methods

Week 10 – Firewalls

- Mon: Windows host-based firewall
- Wed: Linux host-based firewall (iptables/ufw)
- Fri: pfSense firewall overview

Week 11 – IDPS & Wireless Security

- Mon: IDS/IPS concepts & detection
- Wed: IDPS (Snort, Suricata)
- Fri: Wireless network security (WPA2/WPA3)

Week 12 – VPNs

- Mon: VPN concepts & protocols
- Wed: VPN in Windows
- Fri: VPN in Linux

Week 13 – Web Security

- Mon: Web threats (XSS, SQLi, CSRF)
- Wed: Secure coding practices
- Fri: WAFs and secure web development

Week 14 – Network Security Management

- Mon: Security policies, risk management

- Wed: Security audits, compliance
- Fri: Incident response, disaster recovery

Week 15 – Review & Final Prep

- Mon: Comprehensive review
- Wed: Student presentations & demos
- Fri: Final Exam / Project Submission

Assessments & Grading

Component	Weight
Homework	40%
Attendance	5%
Quizzes	15%
Midterm Exam	20%
Final Exam	20%

Grading Scale:

A+: 98–100 | A: 93–97 | A–: 90–92
 B+: 87–89 | B: 83–86 | B–: 80–82
 C+: 77–79 | C: 73–76 | C–: 70–72
 D+: 67–69 | D: 63–66 | D–: 60–62
 F: < 60

Instructor and TA/PLA communication: You may email the Instructor, TA, or PLA any time. Please keep all emails professional. Allow the Instructors and assistant enough time to respond to your email. You can also reach the Instructor, TA, or PLA through the discord channel. Refer to the Discord policy on the content that is allowed on Discord. You may contact the Instructor and assistants in person during their appointed office hours.

Discord Policy: Discord is to be used for assistance during the course. If you have any questions about the course material or homework you are allowed to address it on discord. Discord is not a place to discuss previous graded assignments or any complaints about the course. You may assist fellow students on discord with understanding of course material and homework. You are not allowed to give solutions

in discord. Any form of cheating for any assignments will be referred to the University Cheating policy. Any Violation of the discord policy may result in students being ban from the discord and the student will have to rely on email and office hours to talk to instructors and assistants.

Email Communication:

When emailing the instructor or PLA:

1. Use the subject line to indicate which course and section you are in. This course CS or INFOTC 3810.
2. Be concise and clear about your issue(s). Describe your issue(s) fully. This will help us help you.
3. Make sure you have your full name in the email.

WEB ASSIST: There is a Canvas site dedicated to this class. All grades, reading assignments, homework, and other class materials are hosted on this site. Students are expected to access this site regularly throughout each week, and will require a reliable source of internet access to participate in this course.

COMPUTER PROBLEMS: You must be able to access the Internet and use email for the duration of this course. You must have the software that will be used in this course installed on your computer before each Unit starts. Any technical problems that you may encounter with your computer must be addressed immediately. If the issue cannot be resolved, then a functioning replacement computer should be used. If you miss assigned work due to technical problems or not having the software installed, it will not be considered an “excused absence”. The only exceptions to this are if there has been a campus-wide problem with MU’s email, there is an issue with OneDrive, or there an issue with the Canvas system. The instructor will notify you with an announcement and/or email if this is the case.

LAB COMPUTERS AND SOFTWARE PROFICIENCY: IT students are required to have access to, or otherwise own, a laptop or other computer. Students will also need to acquire software to complete the course. A basic list is provided below. The first week of instruction will be dedicated to making sure students have the correct tools to complete all assignments.

- Web Browser (Firefox, Google Chrome, Microsoft Edge)
- A text editor (VS Code, Notepad++, etc.)

Penalties and Deadline for Lateness

Late assignments will receive a 10% deduction per day of the score (up to 3 days or the equivalent of 30%). Exams and quizzes must be completed by the due date/time to be considered for credit. Submitted work must satisfy all the requirements mentioned for that assignment in order to receive up to full credit.